



MANIPAL

ACADEMY *of* HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5103 Linux OS and Scripting – 2_Theory

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MANIPAL SCHOOL OF INFORMATION SCIENCES

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Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400 (r--)	Try to read and modify the file	<i>Possible to read but can't save the file after writing</i>	Try to list and enter the directory	<i>Can list or see file details with ls -l, but can't enter using 'cd'</i>
200 (-w-)	Try to read and modify the file	<i>Reading permission denied. we can write to the file but the contents of the file are not shown.</i>	Try to list, enter, and create a file	<i>Can't list with ls, permission denied. Can't create a file, need both write and execute permission on the directory to do that</i>
100 (--x)	Try to read and execute the file	<i>Can't read the file and can't execute. Although since sample.txt is not an executable file, it gives 'Exec format error' (Exit code is 126)</i>	Try to list and enter the directory	<i>Can't list contents, but cd into it succeeds (because execute allows traversal). Once inside, ls still fails without read</i>
500 (r-x)	Try to read and modify the file	<i>Can read the file; can't modify it (Permission denied on write)</i>	Try to list and enter the directory	<i>Can list and enter the directory fully. Read and execute permission allows traversal. can't create or delete files</i>
600 (rw-)	Try to read and modify the file	<i>Can read and modify (write) the file successfully</i>	Try to list, enter, and create a file	<i>Can list contents (ls works), but can't cd in or create files</i>
700 (rwx)	Try to read, modify, and execute the file	<i>read, modify, and execute works on the file. Since the file is a .txt a format error occurs</i>	Try to list, enter, create, and delete files	<i>Full access - can list, enter, create, and delete files inside</i>

Screenshots:

400 (r--)

```
haris@ubuntu-arm: ~/Developer/LoS/Theory
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -lR
.:
total 0
-rwxr-xr-x 1 haris haris 19 Aug 22 18:56 sample.txt
drwxr-xr-x 2 haris haris 4096 Aug 22 18:57 sample_dir
./sample_dir:
total 4
-rw-rw-r-- 1 haris haris 36 Aug 22 18:57 anothersample.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$ chmod 400 sample.txt sample_dir/
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
something inside!!
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -l sample_dir/
total 0
-????????? ? ? ? ? ? anothersample.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cd sample_dir/
bash: cd: sample_dir/: Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$
```

200 (-w-)

```
haris@ubuntu-arm: ~/Developer/LoS/Theory
haris@ubuntu-arm: ~/Developer/LoS/Theory$ chmod 200 sample.txt sample_dir/
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
cat: sample.txt: Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat >> sample.txt
another lineharis@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
cat: sample.txt: Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -l sample_dir/
ls: cannot open directory 'sample_dir/': Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ touch sample_dir/newfile.txt
touch: setting times of 'sample_dir/newfile.txt': Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$
```

100 (--x)

```
haris@ubuntu-arm: ~/Developer/LoS/Theory
haris@ubuntu-arm: ~/Developer/LoS/Theory$ chmod 100 sample.txt sample_dir/
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
cat: sample.txt: Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ./sample
bash: ./sample: No such file or directory
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ./sample.txt
./usr/bin/bash: ./sample.txt: Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ echo $?
126
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -l sample_dir/
ls: cannot open directory 'sample_dir/': Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ touch sample_dir/newfile.txt
touch: cannot touch 'sample_dir/newfile.txt': Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$
```

500 (r-x)

```
haris@ubuntu-arm: ~/Developer/LoS/Theory
haris@ubuntu-arm: ~/Developer/LoS/Theory$ chmod 500 sample.txt sample_dir/
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
something inside!!
another lineharis@ubuntu-arm: ~/Developer/LoS/Theory$ cat >> sample.txt
bash: sample.txt: Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -l
total 0
-rwxr-xr-x 1 haris haris 31 Aug 22 18:59 sample.txt
dr-xr-xr-x 2 haris haris 4096 Aug 22 18:57 sample_dir
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -l sample_dir/
total 4
-rw-rw-r-- 1 haris haris 36 Aug 22 18:57 anothersample.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample_dir/anothersample.txt
some other text in some other file
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat >> sample_dir/anothersample.txt
test line 2
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample_dir/anothersample.txt
some other text in some other file
test line 2
haris@ubuntu-arm: ~/Developer/LoS/Theory$ touch sample_dir/newfile.txt
touch: cannot touch 'sample_dir/newfile.txt': Permission denied
haris@ubuntu-arm: ~/Developer/LoS/Theory$
```

A screenshot of a terminal window titled "Aug 22 19:07" and "haris@ubuntu-arm: ~/Developer/LoS/Theory". The terminal shows a series of commands and their outputs:

```
haris@ubuntu-arm:~/Developer/LoS/Theory$ chmod 600 sample.txt; cat sample.txt  
something inside!!  
another line  
haris@ubuntu-arm:~/Developer/LoS/Theory$ ./sample.txt  
bash: ./sample.txt: Permission denied  
haris@ubuntu-arm:~/Developer/LoS/Theory$ echo $?  
126  
haris@ubuntu-arm:~/Developer/LoS/Theory$ ls -l sample_dir/  
total 0  
-?????rwxr-xr-x ? another.sample.txt  
haris@ubuntu-arm:~/Developer/LoS/Theory$ cat sample_dir/another.sample.txt  
cat: sample_dir/another.sample.txt: Permission denied  
haris@ubuntu-arm:~/Developer/LoS/Theory$ nano sample_dir/another.sample.txt  
haris@ubuntu-arm:~/Developer/LoS/Theory$ echo "FILE NOT ACCESSIBLE"  
FILE NOT ACCESSIBLE  
haris@ubuntu-arm:~/Developer/LoS/Theory$ cat >> sample_dir/another.sample.txt  
bash: sample_dir/another.sample.txt: Permission denied  
haris@ubuntu-arm:~/Developer/LoS/Theory$
```


The terminal interface includes standard Ubuntu icons on the left and window controls on the right.

```

Aug 22 19:09
haris@ubuntu-arm: ~/Developer/LoS/Theory
haris@ubuntu-arm: ~/Developer/LoS/Theory$ chmod 700 sample.txt sample_dir/
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
something inside!!
another line
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat >> sample.txt
3rd line
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample.txt
something inside!!
another line
3rd line
haris@ubuntu-arm: ~/Developer/LoS/Theory$ cat sample_dir/anothersample.txt
some other text in some other file
test line 2
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -l sample_dir/
total 4
-rw-rw-r-- 1 haris haris 48 Aug 22 19:04 anothersample.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -lR
.:
total 8
-rwx----- 1 haris haris 40 Aug 22 19:08 sample.txt
drwx----- 2 haris haris 4096 Aug 22 18:57 sample_dir
./sample_dir:
total 4
-rw-rw-r-- 1 haris haris 48 Aug 22 19:04 anothersample.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$ touch sample_dir/newfile.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$ ls -lR
.:
total 8
-rwx----- 1 haris haris 40 Aug 22 19:08 sample.txt
drwx----- 2 haris haris 4096 Aug 22 19:09 sample_dir
./sample_dir:
total 4
-rw-rw-r-- 1 haris haris 48 Aug 22 19:04 anothersample.txt
-rw-rw-r-- 1 haris haris 0 Aug 22 19:09 newfile.txt
haris@ubuntu-arm: ~/Developer/LoS/Theory$

```

1. What does read (r) permission allow you to do with a regular file?
 - View/open the file's contents (with cat), but not modify or run it.
2. What does read (r) permission allow you to do with a directory?
 - List the names of entries inside it (using ls). It does not let us cd into it or see metadata of the entries because that needs execute permission.
3. What does write (w) permission mean for a regular file?
 - Permission to modify, overwrite, or truncate the file's contents (including deleting its contents). No need of read permission
4. What does write (w) permission mean for a directory?
 - Permission to change the directory's contents listing, i.e., create, delete, or rename files inside it. It does not mean we can edit the contents of files inside it.
5. What does execute (x) permission mean for a regular file?

- Permission to run the file as a program/script. If the file isn't a valid executable, it fails with an "Exec format error," and not "Permission denied".
6. Why is execute (x) permission important for accessing a directory?
 - Execute on a directory grants traversal permission - the ability to cd into it, access files inside it by name, and resolve paths through it. Without it, even knowing a file's exact path inside that directory won't let us reach it.
 7. Which permission combination allows the owner to fully access a regular file?
 - read, modify, and execute (**rwX - 700**)
 8. Which permission combination allows the owner to fully access a directory?
 - list, enter, create, and delete files inside (**rwx - 700**)

Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

- A directory isn't a data file in the usual sense - it's more like a lookup table that maps filenames to inodes. That's why the same three permission bits end up meaning something different there.
 - **Read (r)** on a directory only lets us see that mapping, meaning the filenames themselves. It tells us nothing about what's actually inside those files.
 - **Execute (x)** on a directory means we're allowed to traverse or search through it. It has nothing to do with "running" the directory as a program, since directories can't be executed at all.
 - **Write (w)** on a directory controls the mapping itself, so it governs whether we can add or remove entries. It says nothing about whether we can edit the content of the files those entries point to.

In a nut shell, a regular file works differently, since its permissions apply directly to the actual bytes stored in it. Read lets us view that content, write lets us change it, and execute lets us run it as a program.